# AJINKYA AWARI

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## PROFILE

MSc Artificial Intelligence for Biomedicine and Healthcare student at UCL, building reproducible biomedical AI and evaluation systems with explicit evidence and limitations, and reporting weak and null results openly.

## EDUCATION

**University College London (UCL)**, London, UK ***2026-2027***
MSc - Master of Science, Artificial Intelligence for Biomedicine and Healthcare

Savitribai Phule Pune University, Pune, IN | **CGPA: 8.25/10** ***2019-2023***
BE - Bachelor of Engineering, Computer Science

## SKILLS

– **Languages:** Python, C, C++
– **ML / DL Frameworks:** PyTorch, TensorFlow, Keras, Scikit-learn
– **AI & Research Areas:** Graph Neural Networks, Protein Language Models, Federated Learning, Model Evaluation, Explainable AI, Reproducible Research
– **Data Science Tools and Technologies:** Hadoop, Apache Spark, SQL, BigQuery, Docker, Matplotlib
– **Web & Tools:** HTML, CSS, Bootstrap, Flask, Streamlit, Git, AWS, GCP

## EXPERIENCE

**Byju's Think & Learn Pvt. Ltd, India** ***July 2023 – July 2024***
***Cloud Engineer***

– Optimised cloud-based lead management automation by integrating LeadSquared, reducing manual effort by 4+ hours per week and boosting lead conversion rates by 10%, contributing to $300K+ in additional revenue. Utilised cloud monitoring and performance metrics to refine iterations of infrastructure and automation.
– Designed and optimised a scalable cloud infrastructure capable of handling 300,000+ concurrent users by improving backend architecture, database queries, and load balancing, reducing latency by 15% and ensuring 99.7% uptime during peak enrolments.
– Implemented a predictive analytics pipeline on cloud platforms using Python, enabling sales teams to prioritise high-potential leads, improving lead-to-customer conversion by 12% and driving an 8% increase in quarterly revenue.
– Automated cloud-based user onboarding workflows through Python-powered CRM integration, reducing onboarding time by 25%, minimising errors by 30%, and enhancing user satisfaction by 15%, leading to lower churn rates.

**The Leadership 30** ***July 2024 – August 2026***
***Senior Data Analyst***

– Transforming Education in Maharashtra (SDG 4 - Quality Education): Led a data-driven initiative with the Maharashtra government on a three-month project to assess learning gaps in public schools. Developed targeted intervention strategies that boosted student engagement by 15% and improved literacy outcomes for 2,000+ students in rural districts.
– Advancing Clean Energy Access (SDG 7 - Affordable and Clean Energy): Designed a cost-benefit analysis model for renewable energy projects in rural Maharashtra, facilitating policy recommendations for solar micro grids. The initiative led to a 10% increase in clean energy adoption, benefiting 800+ households and reducing annual carbon emissions by 1,500 tons.
– Disaster Resilience & AI-Driven Risk Mitigation: Built an AI-powered flood forecasting system in collaboration with local agencies, enhancing emergency response times by 12% and safeguarding many individuals in 6 flood-prone districts across Maharashtra.

## PROJECTS

**AIXI-Complexity Benchmark**

– Built a local CPU benchmark comparing five decision-making agents (a random policy, tabular Q-learning, an AIXI-inspired planner, a small local language model and an exact dynamic-programming oracle) on 20 frozen finite-state tasks, with 10 replicates per task and fixed seeds.
– Evaluated agents with a normalised oracle-gap metric, a 10,000-resample paired bootstrap per task and hierarchical bootstrap intervals for the suite, supported by 81 tests, a clean lint run and passing continuous integration.
– Applied Holm correction across difficulty proxies and reported the null result openly: no proxy association remained significant (minimum adjusted $p = 0.144$), and the language-model arm was valid on only 10 of the 20 tasks.

**GraphGPS-Upgrade**

– Built a fixture-first graph-level benchmark (graph batching, graph-local positional features and uniform PyTorch Geometric baselines) to compare GraphGPS, GCN and GIN fairly on the official ogbg-molhiv scaffold split.

– Ran a full-budget ablation on a Tesla T4 under one frozen recipe (30 epochs maximum, patience 5, three seeds), giving test ROC-AUC of 0.748 for GraphGPS against 0.698 for GIN and 0.693 for GCN.

– Limited the claim to this frozen recipe and recorded the blocked and one-epoch smoke runs separately, rather than presenting them as benchmark results.

**ESM2 Protein Fitness Benchmark**

– Built a reproducible, leakage-aware benchmark comparing frozen ESM1v and ESM2 representations against simple baselines on ProteinGym single-substitution assays, grouping splits by UniProt ID and sequence-similarity clusters with SHA-256-hashed split manifests.

– Evaluated 40 assays on CPU: ESM1v (five checkpoints, masked-marginal scoring) reached a macro Spearman of 0.475, against 0.266 for frozen ESM2-650M embeddings with ridge regression.

– Covered the pipeline with 112 tests and a privacy-gated public results boundary; grouped bootstrap confidence intervals remain pending GPU access.

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## PUBLICATIONS

V. Paithankar, A. Awari (second author), A. Raskar, S. Patil, N. Jamdar. *Plant Disease Detection using Machine Learning.* IJARSCT, Vol. 3, Issue 2 (doi:10.48175/IJARSCT-9156) and Issue 4 (doi:10.48175/IJARSCT-9297), April 2023.